Novel Tendon Driven Kresling Origami Actuator

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TABLE I: Prototype Maximum Retraction Length (mm)

Stacked Segments	1	2	3	4
Full Extension (Resting State)	36	67	100	129
Size after Full Retraction	23	43	65	87
Percent of Full Size	64	64	65	67

***Abstract*—This work presents a novel modular soft actuator inspired by the Kresling origami fold, designed to achieve compact, scalable, and programmable motion through tendon-driven actuation. The actuator comprises stacked origami units laser-cut from polypropylene sheets, with embedded tendons routed through each segment and controlled via stepper motors. This structure enables significant vertical compression—up to 36% of its original height—while maintaining minimal lateral expansion, a feature advantageous for confined or adaptive robotic applications. The modular design allows for customizable motion profiles and extended actuation range through segment stacking. Despite limitations in torque due to initial motor selection, the actuator demonstrated consistent performance in both vertical compression and directional bending. This platform offers a promising foundation for soft robotic systems requiring reconfigurable geometries, compact form factors, and precise control, with future potential for integration with high-torque motors, alternative materials, and hybrid actuation schemes.**

I. Results

A. Vertical Compression

The following section details the results and properties of the constructed prototype. The design consists of modular Kresling sections made from 0.5mm sheets of laser cut polypropylene which can be stacked on top of each other. Tendons running through the actuator are spooled up and down by 5v 28BYJ-48 stepper motors, which control the motion of the device. In practice however, these motors proved to be underpowered and while control was possible it was limited by the use of these motors. The NEMA17 stepper motor might have been a more appropriate choice in this instance due to a significantly higher torque.

Table I shows the effective range of vertical compression of the stacked Kresling units. The are able to be compressed by approximately one third of their total size which remains roughly consistent across the number of segments. Strong and consistent values relating to vertical compression along its z axis demonstrates the success of the prototype in its key selling point as an actuator capable of compressing its own volume. Alongside minimising its total size, a further benefit is that it is able to reach points more easily that more standard rigid body actuators by compressing its own volume, where a rigid actuator would rely on extending its joints sideways to lower the vertical position of the end-effector.

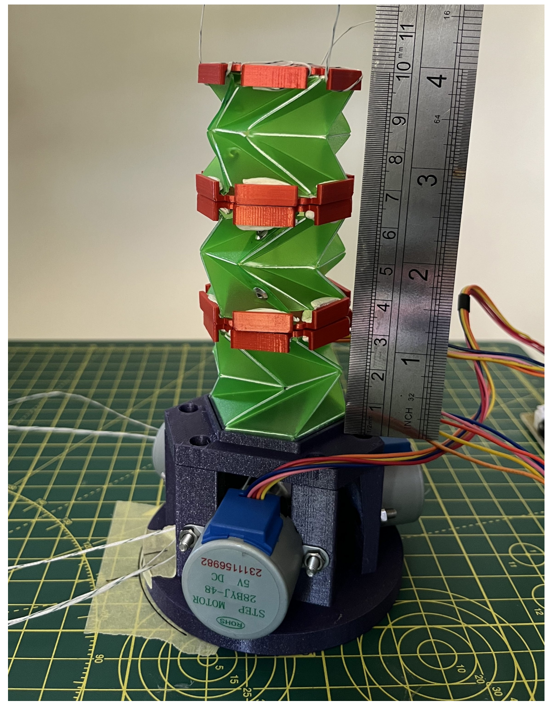


Fig. 1: Experiment Setup with 3 Kresling Sections

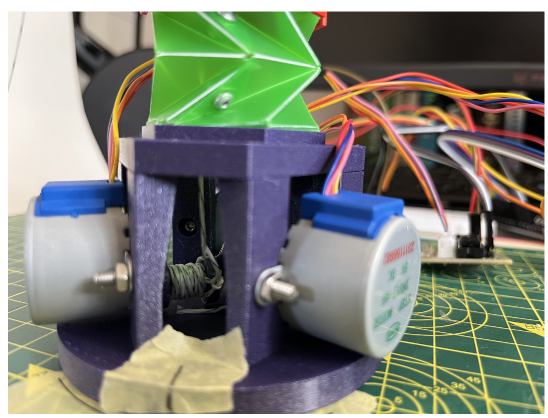
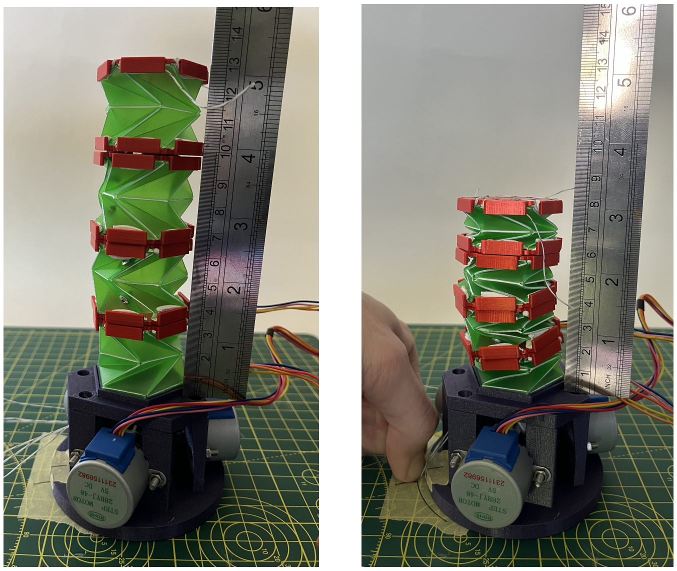


Fig. 2: Tendon Spool

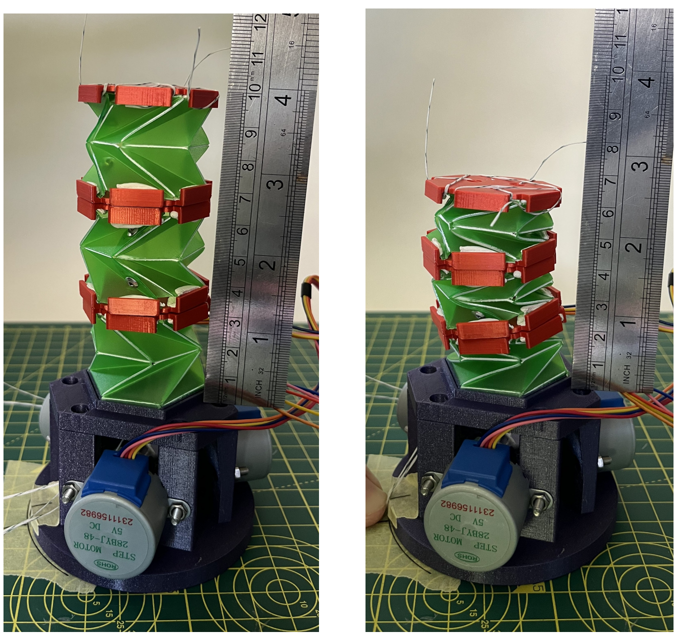
TABLE II: Horizontal bend length (mm) per segmnent

Segments	1	2	3	4
Horizontal Bend	3	11	28	36



(a) At rest/ full extension (b) At full vertical compression

Fig. 3: Experiment showing full vertical compression using 4 Kresling segments

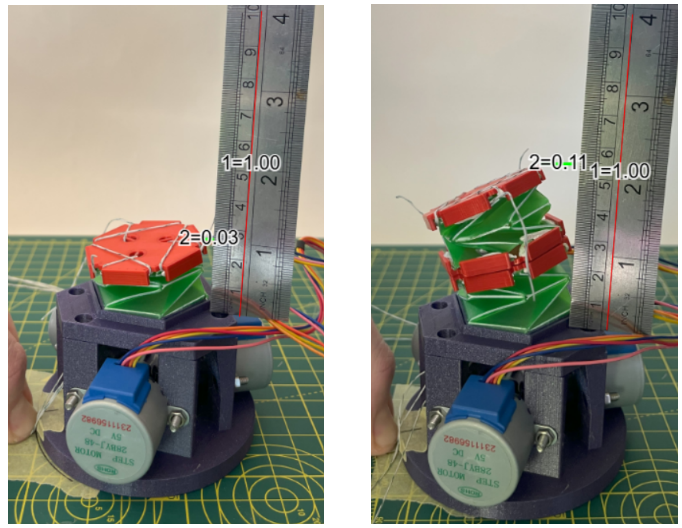


(a) At rest/ full extension (b) At full vertical compression

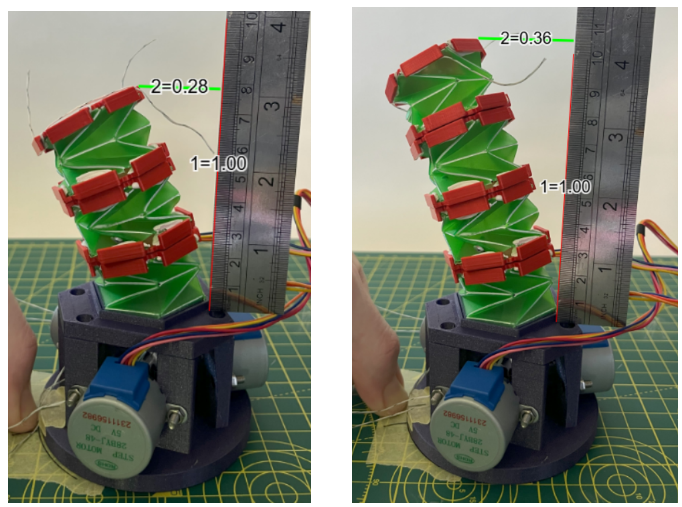
Fig. 4: Experiment showing full vertical compression using 3 Kresling segments

B. Motor Control

As previously mentioned, the chosen motors were overall underpowered and so were not able to provide the full range of motion seen in the results. Despite this control was still possible and effective utilising the motors.



(a) 1 Kresling Core (b) 2 Kresling Cores

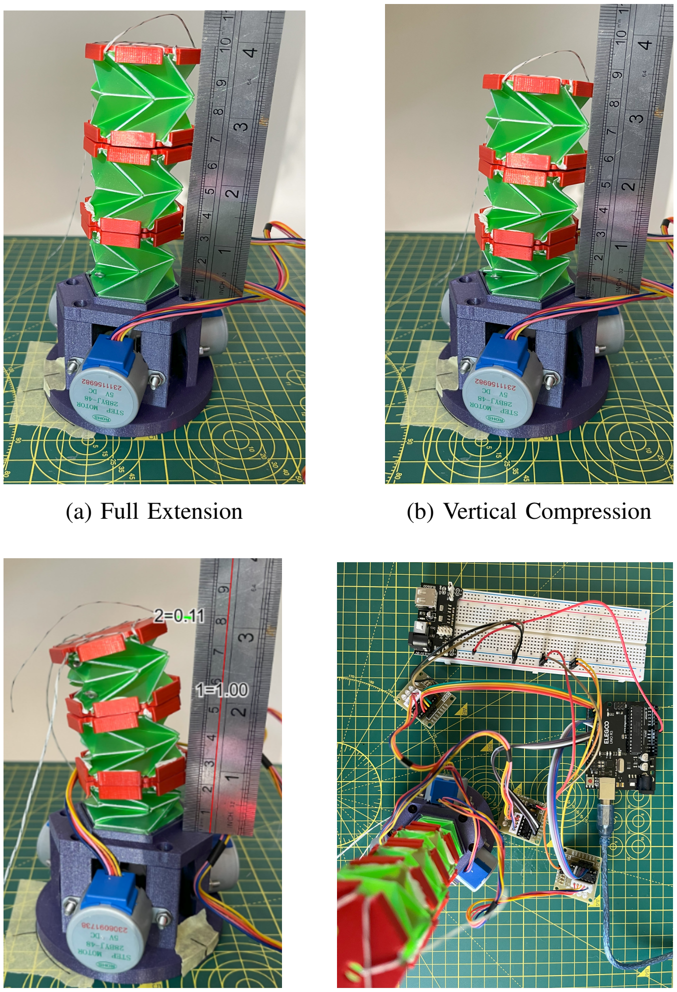


(c) 3 Kresling Cores (d) 4 Kresling Cores

Fig. 5: Comparison of the horizontal bending motion of the actuator

C. Conclusion

This study successfully demonstrated the design and implementation of a modular soft actuator based on the Kresling origami fold, actuated via tendon-driven mechanisms. The actuator exhibited reliable vertical compression and directional bending, validating the effectiveness of its geometric design and modular construction. Key strengths include its ability to minimize horizontal displacement during vertical compression and the ease with which additional segments can be added to scale motion range. While the initial motor choice limited actuation force, the system still achieved meaningful motion control, confirming its functional viability. These results suggest strong potential for further development, particularly through the use of higher-torque motors, advanced materials, and hybrid actuation strategies involving shape memory alloys



(a) Full Extension

(b) Vertical Compression

(c) Horizontal Motion

(d) Circuit

Fig. 6: Circuit and resultant actuator motion

or pneumatics. Overall, this actuator provides a promising platform for soft robotic systems requiring compact, adaptable, and programmable motion, with potential applications in fields ranging from wearable robotics to minimally invasive tools.